

Companion reference — Volume 2

Industrial Network Protocols

Organised by the job they do, not by their name

HOW TO USE THIS

- Not a book to read through — start from the “I want to...” table on page 1
- Every entry answers the same four questions in the same order
- What problem it solves, how it works, when to use it, when not to
- 36 protocol families across 15 sections, one page each

Companion to *Understanding Industrial Networking*. Read that volume first for the concepts; use this one to look things up afterwards.

Generated 21 August 2026.

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By The Job

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HOW TO USE THIS

Find It By The Job

This volume is a lookup tool, not a book to read through. Every entry answers the same four questions in the same order: what problem it solves, how it works, when to use it, when not to. Start from the table below.

I want to...

Goal	Protocol	Page
Stop a spare cable from killing my network	RSTP	2.1
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Lose nothing at all when a cable breaks	PRP, HSR	4.1, 4.2
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Goal	Protocol	Page
Send data to a control centre	DNP3, IEC 60870-5-104	14.5
Send plant data to a server or cloud	OPC UA, MQTT	14.7

SECTION 1

Getting Frames Delivered

1.1 Ethernet

The problem. Many devices share one network and each needs its data to arrive intact and at the right place.

How it works. Data is wrapped in a frame carrying a destination MAC address, a source MAC address, an optional VLAN tag, a type field saying what is inside, the payload, and a checksum. Switches read the destination and forward accordingly. A corrupted frame fails the checksum and is silently discarded — which is why rising error counters, not missing data, are the first sign of a cable problem.

Standard	Speed	Typical medium
100BASE-TX	100 Mbps	Cat 5e copper, older field devices
1000BASE-T	1 Gbps	Cat 6 copper, standard today
1000BASE-SX / LX	1 Gbps	Multimode / singlemode fibre
10GBASE-SR / LR	10 Gbps	Fibre, switch-to-switch

DUPLEX MISMATCH

Set both ends of a link to auto-negotiate, or both to fixed. Never one of each — the auto end cannot detect duplex, falls back to half, and you get a link that works but drops packets under load. The signature is late collisions and rising input errors.

1.2 ARP

The problem. You configure IP addresses, but switches forward using MAC addresses. Something has to connect the two.

How it works. A device that knows an IP but not the MAC behind it broadcasts "who has this address?" The owner replies with its MAC, and the answer is cached for a few minutes.

Worth knowing. ARP is a broadcast, which is why loops produce storms rather than merely duplicate traffic. A **gratuitous ARP** is how a device announces that an address has moved to a new MAC — it is what makes VRRP and redundant controller failover work. If failover completes but traffic does not resume for thirty seconds, suspect that the gratuitous ARP was missed.

1.3 VLANs and 802.1Q

The problem. One switch is one broadcast domain. Everything hears everything, and anyone plugging into any port can reach anything.

How it works. Each port is assigned a VLAN number. Ports in different VLANs behave as if they are on physically separate switches. When a frame crosses a trunk between switches, a four-byte 802.1Q tag is inserted carrying the VLAN ID and a three-bit priority field used by QoS.

Term	Meaning
Access port	Belongs to one VLAN, connects an end device, frames are untagged
Trunk port	Carries several VLANs between switches, frames are tagged
Native VLAN	The one VLAN sent untagged on a trunk — must match at both ends
Allowed list	Which VLANs a trunk may carry — always set this explicitly
QinQ (802.1ad)	A second tag wrapping the first, for carrying a customer's VLANs across a provider network

TWO HABITS WORTH ADOPTING

Never use VLAN 1. It is the default on every vendor's equipment and cannot be deleted. Move management to a dedicated VLAN, shut down the VLAN 1 interface, and remove VLAN 1 from trunks.

Set a native VLAN that carries no traffic. If one end says native VLAN 999 and the other says 1, untagged frames land in the wrong VLAN and both switches log continuous mismatch errors.

SECTION 2

Preventing Loops

2.1 STP, RSTP, MSTP

The problem. A redundant path is also a loop, and an Ethernet frame in a loop never expires. One broadcast becomes millions in seconds.

How it works. Switches elect a root, then each switch blocks any port that would complete a loop back to it. Blocked ports have link but carry no data. When the active path fails, a blocked port begins forwarding.

Version	Standard	Recovery	Notes
STP	802.1D	30–50 s	Obsolete. Never specify for new work.
RSTP	802.1w	1–3 s	The default choice. Vendor-neutral.
MSTP	802.1s	1–3 s	Groups VLANs into instances with different roots — lets you use both uplinks
PVST+	Cisco	1–3 s	One instance per VLAN. Heavy with many VLANs.

Use it when. Always, on every managed switch, even where you believe no loop exists. It costs nothing and it is the only thing standing between you and a storm when someone patches a spare lead.

Don't rely on it when. Recovery must be faster than about a second — then you need a ring protocol (Section 3) or PRP/HSR (Section 4).

SET THE ROOT DELIBERATELY

Left alone, switches elect whichever has the lowest MAC address — usually the oldest unit in the least convenient cabinet. Force it to your main switch, and set the second-lowest priority on your backup. Then record which ports block in the healthy state, because a healthy network and one already running on its backup path look identical.

2.2 STP guards

The problem. STP handles loops between switches but does nothing about a switch appearing where an end device should be, or a fibre that fails in one direction only.

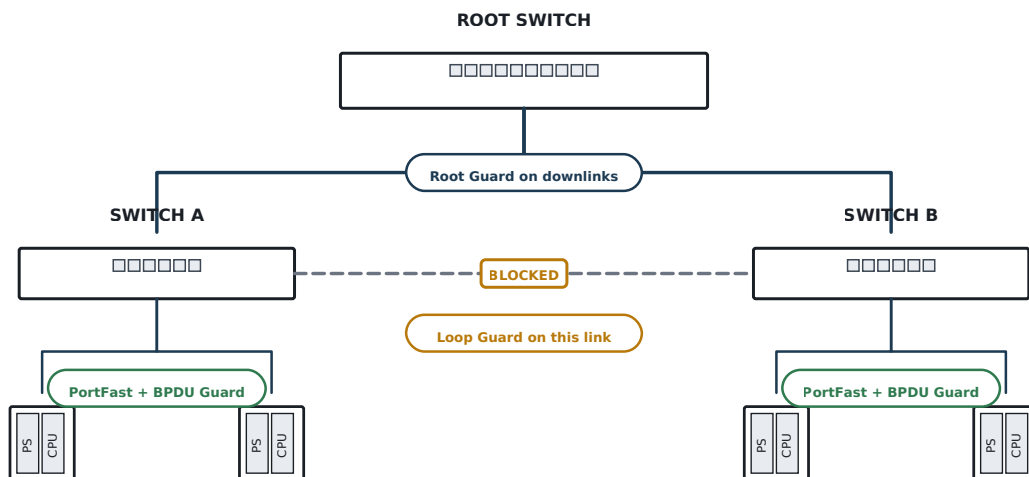


Figure 1. Where each protection belongs. PortFast and BPDU Guard on device ports, Root Guard on downlinks, Loop Guard on links between switches.

Feature	Apply to	Prevents
PortFast	Access ports only	The 30-second delay before a PLC gets link after reboot
BPDU Guard	Access ports only	An unauthorised switch or a loop on a device port
Root Guard	Ports facing downstream switches	A small switch claiming the root role and redirecting all traffic
Loop Guard	Links between switches	A one-way fibre failure causing a blocked port to start forwarding

NEVER PORTFAST AN UPLINK

PortFast makes a port forward immediately without waiting for loop detection. On a link between switches that is exactly how a transient loop becomes a storm. PortFast belongs only where an end device is connected.

SECTION 3

Ring Recovery

3.1 MRP, DLR, REP and vendor rings

The problem. RSTP takes one to three seconds to recover. Some control loops cannot tolerate that.

How it works. A ring protocol knows in advance that the topology is a ring, so it skips the general-purpose discovery that makes RSTP slow. One node is the ring manager. It blocks one port, sends test frames around the ring continuously, and unblocks within milliseconds when they stop arriving.

Protocol	Standard	Recovery	Ecosystem
MRP	IEC 62439-2	200 ms standard 10–30 ms fast	Vendor-neutral, common on PROFINET equipment
DLR	ODVA	~3 ms	EtherNet/IP; devices themselves must support it
REP	Cisco	50–100 ms	Cisco industrial switches
Turbo Ring, HIPER-Ring	Proprietary	10–300 ms	Moxa, Hirschmann and similar

Use it when. Cable naturally runs in a loop — along a machine hall, around a process area — and you need sub-second recovery.

Don't use it when. The equipment is mixed-vendor and you need better than MRP speed. Proprietary ring protocols do not interoperate, so a mixed ring falls back to RSTP or MRP and you inherit the slower figure. Decide this at procurement, not at commissioning.

SIZING RULE

Maximum tolerable recovery = tightest application timeout ÷ 3. A loop polled every 500 ms tolerates 200 ms MRP but not 3 s RSTP. Work this out per traffic type before choosing.

SECTION 4

Zero-Loss Redundancy

4.1 PRP

The problem. Even 10 ms of recovery loses data. Electrical protection and motion control cannot accept any loss at all.

How it works. Two completely independent networks, with no connection between them. Every frame is transmitted on both, carrying a sequence number. The receiver accepts the first copy to arrive and discards the second. If one network fails, the second copy simply never arrives — and nothing needs to recover, because the backup path was already carrying live traffic.

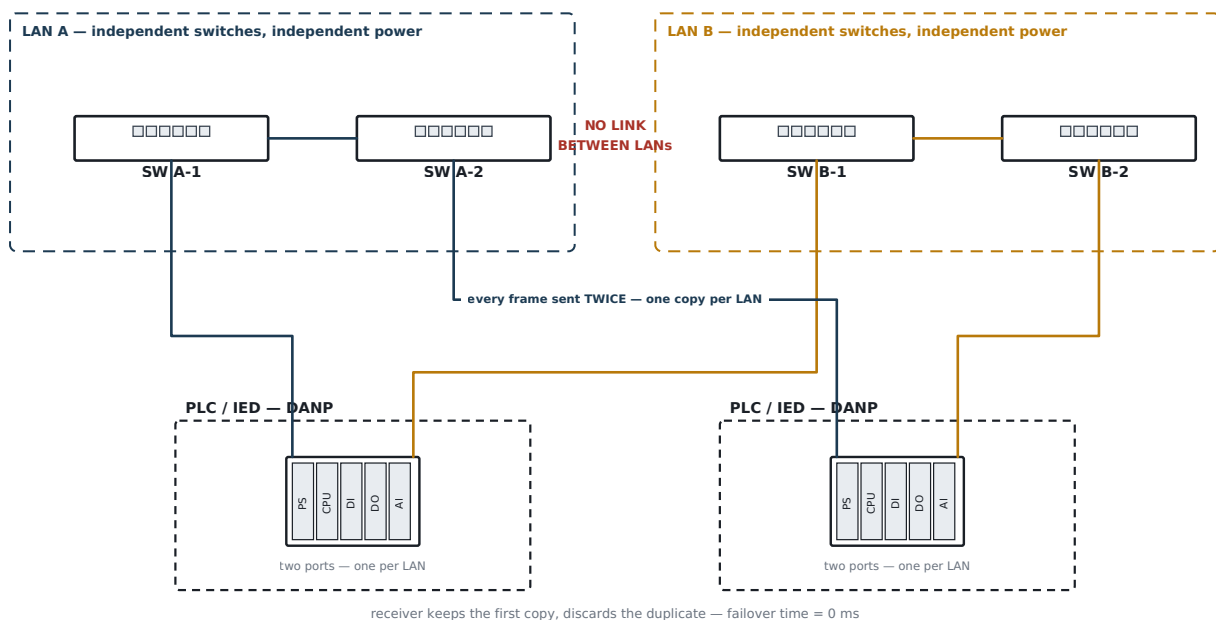


Figure 2. PRP. Two independent LANs, every frame duplicated, no convergence to wait for.

Property	Value
Standard	IEC 62439-3 Clause 4
Recovery time	Zero — there is nothing to converge
Bandwidth cost	Double
Device requirement	Two ports and PRP support, or a RedBox for ordinary devices
Switch requirement	None special — the two LANs are ordinary networks

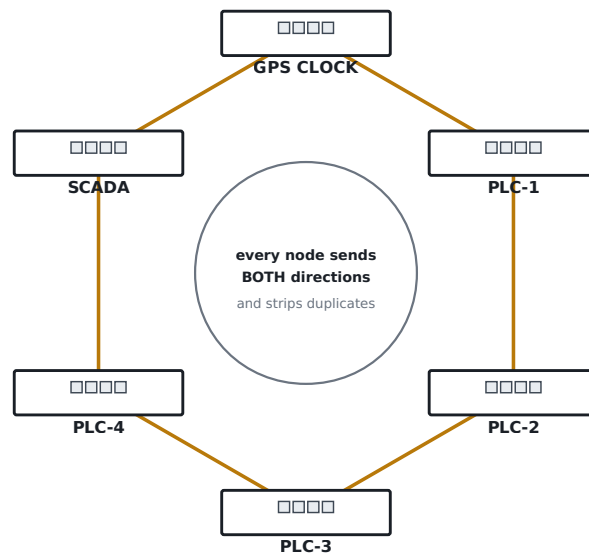
Use it when. No loss is acceptable, devices are spread out, and you can justify two of everything. It tolerates mixed switch vendors because each LAN is a normal network.

Don't use it when. The two LANs would share a cabinet, a power supply, or a cable route — that defeats the entire design and you have paid twice for nothing.

4.2 HSR

The problem. Same as PRP, but PRP needs two complete networks and twice the cabling.

How it works. The same duplicate-frame idea arranged as a ring. Each node injects a copy of every frame in both directions and strips duplicates it has already seen. Recovery is zero, as with PRP, but with one ring instead of two LANs.



zero failover time — but every node must be HSR-capable silicon

Figure 3. HSR. Duplicates travel both ways around the ring and are removed in hardware.

Use it when. Devices are physically close — a single row of panels, a ring around one machine — and cable cost matters.

Don't use it when. Any device in the ring is not HSR-capable. Every node must forward and deduplicate at line rate, which needs purpose-built hardware. A single ordinary switch inserted into the ring breaks it.

OFTEN USED TOGETHER

Large installations run HSR rings at equipment level, coupled into a PRP network at station level through RedBoxes. **802.1CB** generalises the same duplicate-and-eliminate idea into standard Ethernet as part of TSN.

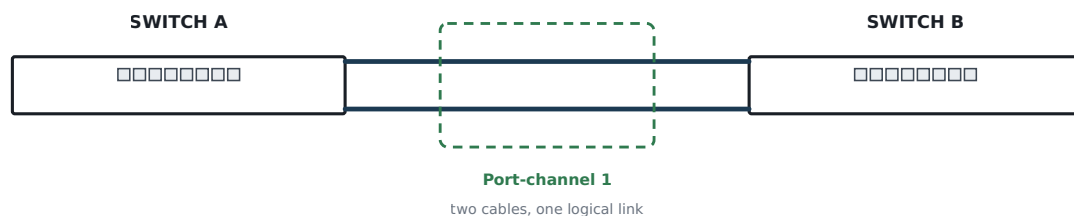
SECTION 5

Using Every Link

5.1 LACP and link aggregation

The problem. Two cables between the same pair of switches is a loop, so STP blocks one. You have paid for two links and can use one.

How it works. Link aggregation bundles several physical links into one logical link. STP sees a single link, so nothing is blocked, and both members carry traffic. If a member fails the bundle stays up with reduced bandwidth and no session drops.



Both links forward at the same time - nothing is blocked.
STP sees one link, so no loop exists. Lose one cable and bandwidth halves,
but the link stays up and no session drops.

Figure 4. Link aggregation. Two cables presented to the rest of the network as one, so both forward.

Method	Standard	Notes
LACP	802.3ad / 802.1AX	Use this. Negotiates with the far end and detects miswiring.
PAgP	Cisco	Cisco-only equivalent
Static ("on")	—	No negotiation. A wiring mistake becomes a loop with no warning.

Use it when. Two switches need more bandwidth or link redundancy between them, and both ends are the same physical switch.

Don't use it when. The two links go to *different* switches — ordinary LACP cannot span two chassis. That needs MLAG (next page).

AGGREGATION DOES NOT MULTIPLY SINGLE-SESSION SPEED

A bundle hashes each conversation onto one member link, so one file transfer between two devices never exceeds the speed of a single member. Two 1 Gbps links do not give one device 2 Gbps. They give the aggregate more capacity and every link a backup.

Check with the summary command: the bundle should show as up, and every member as bundled. A member showing "independent" usually means one end is set to static while the other is set to active.

5.2 MLAG, stacking, VSS and vPC

The problem. In a dual star every access switch has two uplinks to two different core switches. LACP cannot bundle across two chassis, so STP blocks one uplink and half your capacity sits idle.

How it works. The two core switches are joined by a dedicated peer link and present themselves to everything below as a single logical switch. The access switch then forms one ordinary LACP bundle across both uplinks. Nothing blocks, both forward, and losing a whole core switch loses half the bandwidth rather than the link.

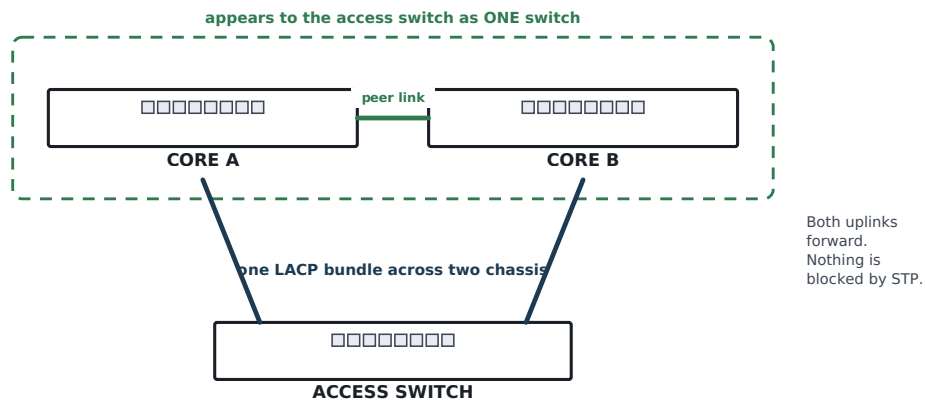


Figure 5. Two core switches acting as one. The access switch bundles both uplinks with ordinary LACP.

Name	Vendor	Notes
MLAG / MC-LAG	Generic term	Switches stay separate devices with separate configs
Stacking	Most vendors	Switches merge into one device with one config — simplest to manage
VSS	Cisco	Two chassis, one control plane
vPC	Cisco Nexus	Separate control planes, shared forwarding

Use it when. You have a dual star and want both uplinks working. This is the cleanest answer.

Don't use it when. You need the two cores to fail completely independently. Stacking in particular merges the control plane, so a software fault or an upgrade can affect both units at once. Where independence matters more than simplicity, keep them separate and use MSTP with split roots instead.

THE SIMPLER ALTERNATIVE

If MLAG is unavailable, run **MSTP** and make Core A the root for half your VLANs and Core B the root for the other half. Each uplink then carries some VLANs and backs up the rest. Not as elegant, no extra hardware, and it works on any managed switch.

SECTION 6

Gateway Redundancy

6.1 VRRP, HSRP and GLBP

The problem. Devices are configured with one gateway address. If that device fails, every host in the subnet is stranded — even with a healthy second router beside it. Hosts do not fail over on their own.

How it works. Two switches share a **virtual IP address** and a virtual MAC. Hosts point at the virtual address. One switch answers for it; the other watches. On failure the standby takes over the virtual address and announces the change with a gratuitous ARP, and hosts notice nothing.

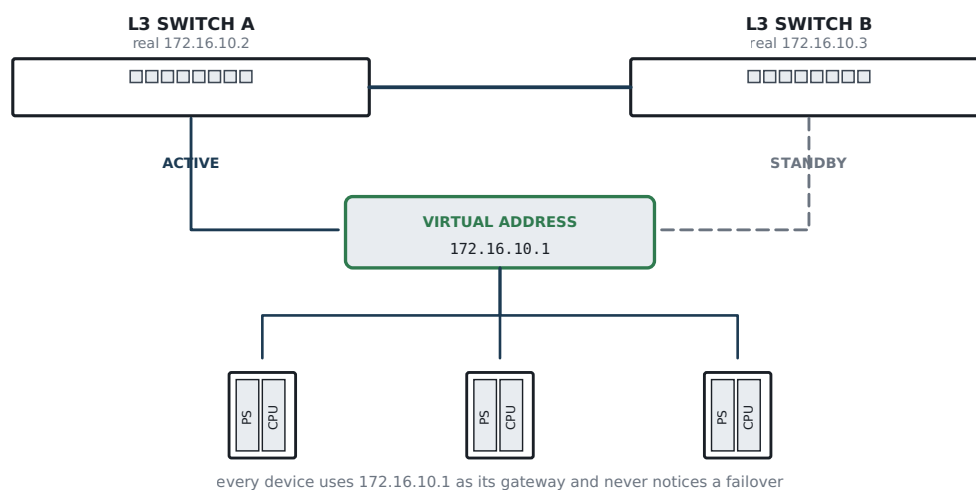


Figure 6. Gateway redundancy. Devices use the virtual address and never need to know which switch is answering.

	VRRP	HSRP	GLBP
Standard	RFC 5798, open	Cisco	Cisco
Mixed vendor	Yes	No	No
Load sharing	Per VLAN group	Per VLAN group	Native, per host
Roles	Master / Backup	Active / Standby	AVG / AVF

Use it when. You have two Layer 3 switches and hosts that need a gateway. In a mixed-vendor plant, choose **VRRP** — it interoperates.

THREE SETTINGS THAT DECIDE WHETHER IT ACTUALLY WORKS

Preempt. Without it, a recovered primary stays passive forever. Your redundancy silently degrades after the first failure and you are unprotected against the second.

Preempt delay. Without it, a rebooting switch claims the active role while its own uplink is still down — creating a black hole.

Uplink tracking. Lowers priority when the upstream link fails, so the role moves. Without it a switch stays active with no path onward.

Split active roles across VLANs so both switches carry traffic instead of one sitting idle.

SECTION 7

Routing

7.1 Static routes, OSPF, EIGRP, BGP

The problem. Traffic must reach a subnet that is not local — another VLAN, another site, a control centre.

How it works. A router or Layer 3 switch holds a table mapping destination networks to next hops. Static routes are typed in by hand. Dynamic protocols let routers learn from each other and re-route automatically when something breaks.

Method	Scale	Convergence	Use when
Static	A handful of routes	None — manual	Most plants. Predictable, cannot be destabilised by a neighbour.
OSPF	Large, open standard	Seconds	Several sites, multiple paths, mixed vendors
EIGRP	Large, Cisco	Fast	All-Cisco networks
RIP	Tiny, obsolete	Very slow	Teaching only
BGP	Internet scale	Slow, deliberate	Between organisations, utility WANs

Use static when. The plant has one or two paths out and the route count is small enough to maintain by hand. Predictability is worth more than automation here.

Use OSPF when. There are several sites with more than one path between them, and hand-maintaining routes has become error-prone.

SCOPE IT TIGHTLY

If you do run OSPF, make every interface passive by default and enable it only on the links where a neighbour genuinely exists, and authenticate the adjacencies. That stops the switch forming relationships on VLANs where no router should be — a stability and a security measure at once.

SECTION 8

Multicast Control

8.1 IGMP and IGMP snooping

The problem. Multicast is addressed to a group rather than a device. A switch that does not understand it treats it like a broadcast and floods every port. One camera or one process-data stream then reaches every PLC on the VLAN, permanently.

How it works. Devices that want a stream send an IGMP join message. A switch with **IGMP snooping** enabled listens to those messages and builds a table of which ports want which group, then forwards each stream only to those ports.

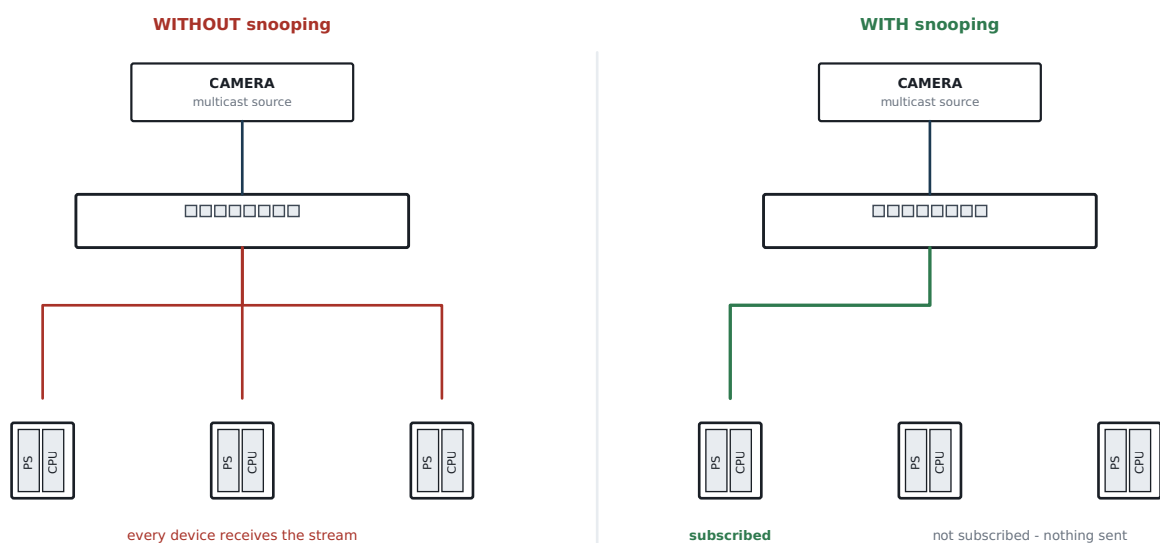


Figure 7. The same network with and without IGMP snooping. Without it, every device processes traffic it never asked for.

Use it when. Any multicast exists on the network — video, process data, or discovery protocols. In practice: nearly always.

SNOOPING NEEDS A QUERIER, AND THIS IS THE CLASSIC MISTAKE

IGMP snooping only works if something on the segment periodically asks "who still wants this?" On a routed network the router does that. On an isolated VLAN with no router, **nothing does** — memberships time out after a couple of minutes and the switch quietly reverts to flooding.

The symptom is unmistakable once you know it: multicast behaves perfectly for the first few minutes after commissioning, then floods. Configure a querier explicitly on every isolated multicast VLAN.

Related. PIM routes multicast between subnets — needed only if multicast must cross a Layer 3 boundary, which in a plant it usually should not. Layer 2 multicast without IP, such as certain protection messages, carries no IGMP at all and must be handled by VLAN separation or static filtering instead.

SECTION 9

Prioritising Traffic

9.1 QoS: 802.1p and DSCP

The problem. When two frames want the same port at once, one waits. Without configuration the switch has no idea which one matters. A backup running to the historian can sit in front of a control command.

How it works. Frames are marked with a priority — three bits in the 802.1Q tag (**802.1p CoS**, values 0–7) or six bits in the IP header (**DSCP**). Each switch port has several output queues, and marked traffic is placed into the matching queue. High-priority queues are emptied first.

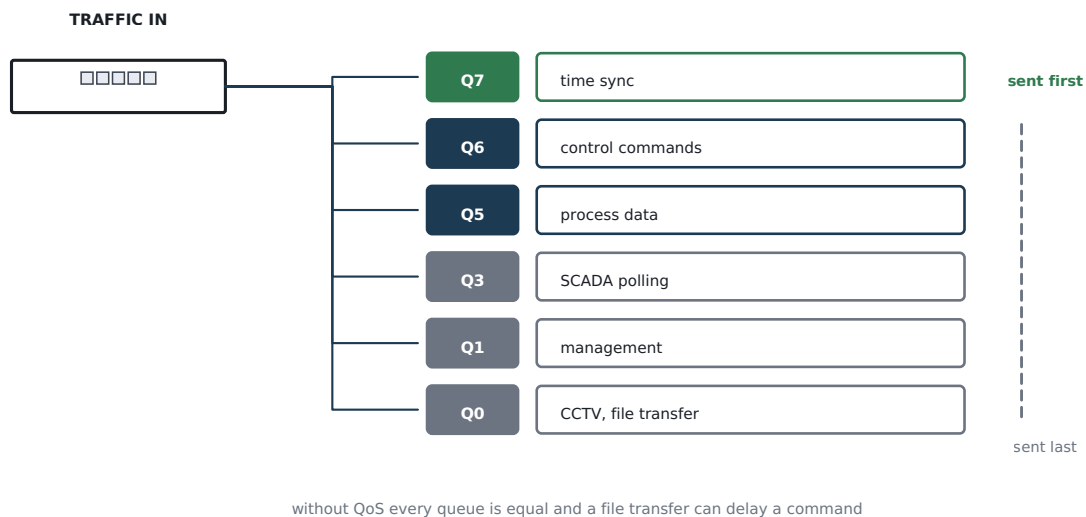


Figure 8. Egress queueing. Time sync and control traffic leave first; bulk traffic waits.

Traffic	Priority	Handling
Time synchronisation	7	Strict priority — delay becomes clock error
Control and trip commands	6	Strict priority
Cyclic process data	5	Strict priority
SCADA polling	3	Weighted
Management (SSH, SNMP)	2	Weighted — must survive congestion for diagnosis
CCTV, file transfer, backups	0	Best effort

TRUST BOUNDARIES

Trusting priority markings on an access port means any device plugged in can mark its own traffic as most important. Trust markings on trunks between switches you control; classify traffic yourself on access ports based on what it actually is.

9.2 TSN

The problem. QoS makes important traffic likely to be fast. Some applications need it *guaranteed* to be fast.

How it works. Time-Sensitive Networking is a set of IEEE 802.1 amendments that add scheduling to Ethernet. All devices share a synchronised clock, and time is divided into slots reserved for specific traffic. Critical traffic has a slot nobody else may use, so its latency is bounded rather than merely probable.

Standard	Provides
802.1AS	The synchronised clock everything else depends on
802.1Qbv	Scheduled traffic — the time slots
802.1Qbu	Frame preemption — a large frame can be interrupted mid-transmission
802.1CB	Frame replication and elimination — the PRP/HSR idea, standardised

Use it when. Designing new, with a long horizon, and motion control or hard real-time requirements.

Don't use it when. Retrofitting. Every device and every switch in the path must support it, and conventional QoS solves most plant problems adequately.

SECTION 10

Addressing and Basic Services

10.1 IP addressing and subnets

The problem. MAC addresses have no structure, so they cannot be used to decide where to send traffic that is leaving the local network.

How it works. An IP address is paired with a subnet mask that splits it into a network part and a device part. `172.16.30.15 / 255.255.255.0` means network `172.16.30`, device `15`. Everything sharing the network part is local; anything else goes to the gateway.

Mask	Short form	Usable addresses	Typical use
255.255.255.0	/24	254	Standard choice. One per VLAN.
255.255.255.128	/25	126	Splitting a /24 in half
255.255.255.252	/30	2	Point-to-point link between routers
255.255.0.0	/16	65534	Too large for one broadcast domain

Private ranges — `10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16` — are reserved for internal use and never appear on the public internet. Use one consistently and map the third octet to your VLAN numbers so an address tells you what a device is.

10.2 DHCP, DNS, NAT and ICMP

Protocol	Does	In a plant
DHCP	Hands out addresses automatically	Avoid for control devices. A device rebooting during a disturbance must not depend on a server answering. Use it for engineering laptops on a separate VLAN.
DNS	Turns names into addresses	Useful for servers. Control devices are addressed numerically so they work when DNS does not.
NAT	Rewrites addresses at a boundary	Common where a vendor skid ships with fixed addresses that clash with your plan. Hides the machine behind one address.
ICMP	Carries ping and error messages	Your first diagnostic. Do not block it inside the plant — it costs you troubleshooting for almost no security gain.

10.3 TCP and UDP

	TCP	UDP
Delivery	Guaranteed, retried, in order	Best effort, no retry
Overhead	Connection setup, acknowledgements	Almost none

	TCP	UDP
Latency	A retry costs time	Predictable
Used by	Modbus TCP, MMS, DNP3, OPC UA	Cyclic process data, PTP, NTP, video

The trade-off is that a TCP retry is worse than a lost frame for cyclic data — a stale value delivered late is less useful than the next fresh one. That is why real-time process protocols use UDP or raw Ethernet while request-and-reply protocols use TCP.

SECTION 11

Time

11.1 NTP and SNTP

The problem. Every device has its own clock and they drift apart by seconds a day. Events logged by different devices cannot be placed in order.

How it works. A client asks a server for the time, measures the round trip, and assumes the path is symmetric — that each direction took half. It adjusts its clock gradually rather than stepping it, so time never jumps backwards.

Accuracy. A few milliseconds on a local network. Good enough for logs, alarms and SCADA events. **SNTP** is a simplified version with the same accuracy and less sophisticated filtering; most field devices implement it.

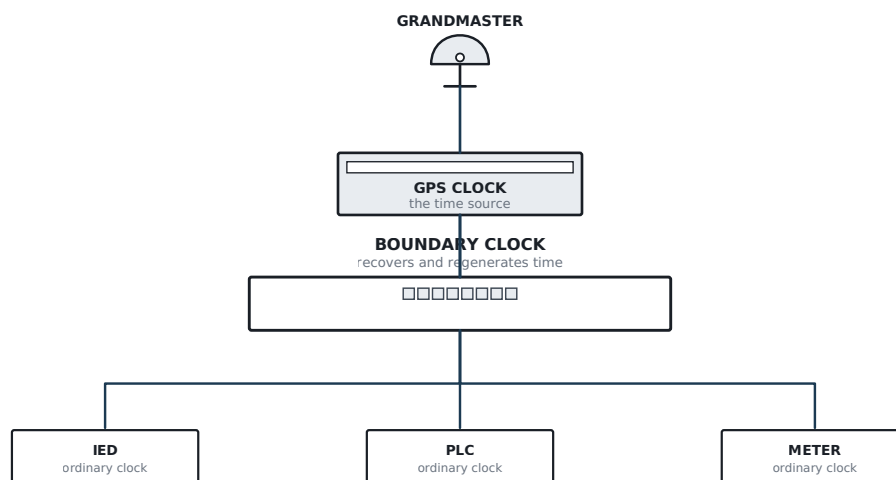
CONFIGURE TWO SOURCES

A single source that fails silently — still answering, but with the wrong time — drags the whole plant with it and nothing alarms. Two or more sources let the client spot the disagreement.

11.2 PTP (IEEE 1588)

The problem. NTP's symmetric-path assumption breaks in a switched network with variable queuing, so it cannot reach better than about a millisecond.

How it works. Two changes. Timestamps are applied by the network hardware as the frame crosses the wire rather than by software, removing the largest source of error. And switches participate: a **boundary clock** recovers time and regenerates it downstream, while a **transparent clock** measures how long each frame spent inside it and writes that into a correction field.



A switch that is not PTP-aware adds unpredictable delay and destroys accuracy - every switch in the path must participate.

Figure 9. A PTP hierarchy. Every switch in the path must participate, or its queuing delay becomes unmeasured error.

Profile	Used for
C37.238 (Power Profile)	Electrical substations and generating plant
802.1AS (gPTP)	TSN and factory automation
Default (Annex F / D)	General purpose, Layer 2 or over IP

TWO WAYS PTP SILENTLY FAILS

An ordinary switch in the path. It adds queuing delay that varies with load, which PTP cannot measure or compensate. The resulting error changes with traffic — worse than a fixed offset, because it cannot be calibrated out. Every switch between source and device must be a boundary or transparent clock.

Profile mismatch. All devices in a domain must use the same profile, the same domain number and the same delay mechanism. A mismatch does not produce an error message — the device simply never leaves the listening state.

11.3 IRIG-B and 1PPS

The problem. Older equipment has no PTP support, and network-based timing depends on the network working.

How it works. A dedicated wire carries a timing signal from a master clock straight to each device. There is no network involved at all, which makes it the most robust option available.

Format	Accuracy	Notes
IRIG-B modulated (B12x)	~10 μ s	Amplitude-modulated on coax, older equipment
IRIG-B unmodulated (B00x)	~1 μ s	DC level shift, better accuracy
1 PPS	< 100 ns	One pulse per second; needs a separate date source

Use it when. Devices support it and you want timing independent of network health. Keeping IRIG-B for legacy equipment is not a compromise — it is the most reliable path those devices have.

THE FAILURE NOBODY NOTICES

A GPS clock that loses satellite lock does not stop. It keeps producing time and drifts — for hours or days depending on its internal oscillator. Nothing alarms unless you configure it to. Wire the clock's alarm contact into SCADA and poll its lock status. Otherwise the first sign is an unusable fault record weeks later.

SECTION 12

Seeing What Is Happening

12.1 LLDP and CDP

The problem. Nobody knows what is plugged into which port, and the drawing is three years out of date.

How it works. Each switch periodically announces its identity, port name and capabilities to its directly connected neighbours. Every device builds a table of what it can see. **LLDP** (802.1AB) is the open standard; **CDP** is Cisco's equivalent.

Use it when. Always on links between switches — it turns "which cable is this?" into a one-command answer and lets monitoring tools draw an accurate map automatically.

Turn it off when. On access ports facing devices you do not control. Announcing your switch model, software version and port name to anything that plugs in is unnecessary exposure.

12.2 SNMP and syslog

The problem. A link failed at three in the morning and nobody knew until production stopped.

	SNMP	Syslog
Direction	Monitoring system polls devices; devices send traps on events	Devices push messages to a collector
Carries	Interface counters, CPU, temperature, errors	Event text: link up/down, config change, security violation
Use for	Trends and thresholds	What happened and when

Versions matter. SNMPv1 and v2c send a community string in plain text and are effectively unauthenticated. Use **SNMPv3**, which authenticates and encrypts. If a device only supports v2c, restrict it to read-only and limit which addresses may query it.

SYSLOG IS ONLY USEFUL WITH CORRECT TIME

Log messages are timestamped by the device that generates them. If clocks disagree, the collector's view of event order is wrong and the investigation is worthless. This is the practical reason Section 11 matters.

Configure both together: point every device at the same time source and the same syslog collector.

12.3 SPAN and NetFlow

The problem. Counters tell you that something is wrong. They do not tell you what the traffic actually contains.

How it works. SPAN (port mirroring) copies traffic from one port or VLAN to another port where a laptop running a capture tool is connected. **NetFlow** summarises rather than copies — it records who talked to whom, on which port, and how much, without storing the payload.

Use SPAN when. Diagnosing a specific fault — a device not responding, a protocol behaving oddly, a suspected storm.

Use NetFlow when. You want a continuous picture of what normally talks to what, which makes anything unusual obvious.

MIRROR CAPACITY

Mirroring a busy gigabit VLAN to a gigabit port drops frames as soon as combined traffic exceeds the port rate — and the frames it drops will not be the ones you were looking for. Mirror one direction only, or filter to specific source ports, when capturing on a busy segment.

SECTION 13

Security

13.1 Port security and 802.1X

The problem. Any port is an open door. Anyone who reaches a panel can plug in a laptop.

Control	How it works	Suits
Port security	Limits how many MAC addresses may appear on a port, and optionally which	Field devices that never move. Simple, no infrastructure needed.
802.1X	Device must authenticate against a RADIUS server before the port opens	Offices and workstations. Needs supplicant support the device may not have.
Disable unused ports	Shut them and put them in a dead VLAN	Everywhere. Free.

CHOOSE THE VIOLATION ACTION CAREFULLY

Shutting a port on violation means an unexpected event takes equipment offline until someone attends. On a control device port that is a self-inflicted outage. **Restrict** — drop the offending traffic, raise an alarm, keep the port up — is almost always the right choice in a plant. Reserve shutdown for ports in publicly accessible areas.

Also: set the MAC limit above one where a redundant controller pair shares a port, or the first changeover triggers a violation.

13.2 ACLs, RADIUS and MACsec

The problem. Devices that can reach each other will, and most industrial protocols have no authentication of their own.

Control	Does	Notes
ACL	Permits or denies traffic by source, destination and port number	The primary protection for Modbus, which has no security of its own. Log the denies and review them.
RADIUS / TACACS+	Central login for switch administration	Removes shared local passwords and gives you an audit trail
MACsec	Encrypts traffic on the wire, hop by hop	For links crossing areas you do not physically control
SSH, HTTPS	Encrypted management access	Disable Telnet and plain HTTP entirely

An ACL is worth more than it looks. A rule permitting only your two SCADA servers to reach port 502 on the field VLAN, denying everything else and logging it, is the single highest-value security control available on a Modbus network. It takes a few lines.

13.3 Firewalls and the IDMZ

The problem. The plant network must exchange data with the business network without being reachable from it.

How it works. A demilitarised zone sits between them. A session from outside terminates in the DMZ; a separate session continues inward. No single connection ever spans both sides.

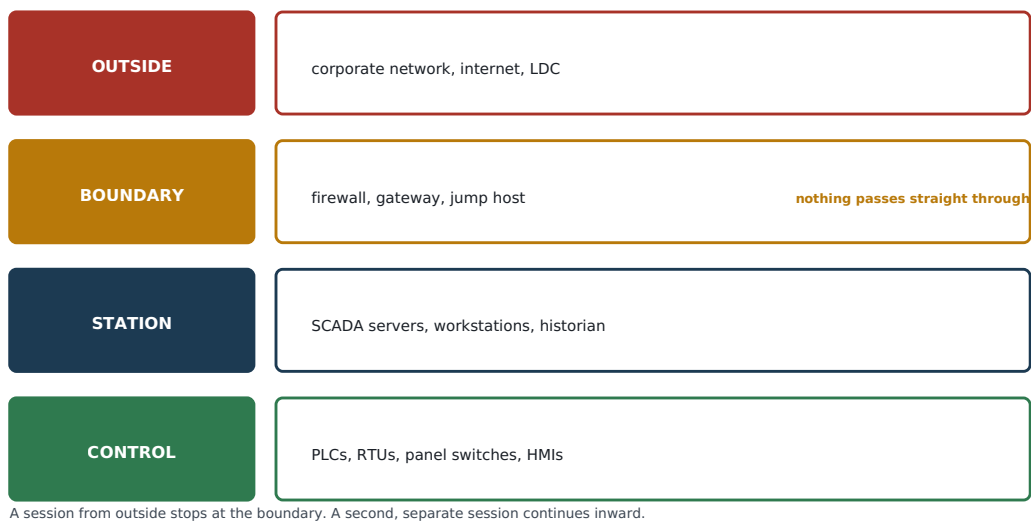


Figure 10. Zones and the boundary between them. Nothing passes straight through from outside to control equipment.

Pattern	How it is done
Historian data to business	Plant historian pushes to a replica in the DMZ; business users query the replica
Remote engineering access	Connect to a jump host in the DMZ with two-factor, then open a second session inward
Patches and antivirus	Downloaded to a DMZ server, validated, then pulled inward on a schedule
Vendor access	Time-limited, brokered through the jump host, recorded, disabled by default

A FIREWALL RULE IS NOT A DMZ

Permitting the business network to reach the plant historian directly — however tightly scoped the rule — still means a compromised business machine has a path to control equipment. The point of the DMZ is that no session crosses; the data does, through an intermediary.

SECTION 14

Field Protocols

14.1 Modbus TCP and Modbus RTU

The problem. Read and write values in field equipment, as simply as possible.

How it works. Data sits in numbered registers. A client opens a connection to **TCP port 502** and requests a range; the device answers. RTU is the same idea over a serial wire, with devices addressed by unit ID on a shared bus.

Register type	Size and access	Typical use
Coil	1 bit, read/write	Start command, digital output
Discrete input	1 bit, read only	Limit switch, status
Input register	16 bit, read only	Measured value
Holding register	16 bit, read/write	Setpoint, general reading

Use it when. Connecting mixed-vendor equipment. Almost everything supports it.

THREE RECURRING PROBLEMS

No security whatsoever. No password, no encryption. Anyone with a path to port 502 can write. Segmentation and ACLs are the only protection.

Inconsistent register numbering. Some manuals count from zero, some from one, and the "4x" convention is applied differently by different manufacturers. Byte order for 32-bit values varies too. Verify against a known live value, not the manual.

Timeouts below network recovery time. If failover takes two seconds and your timeout is one, every normal failover marks devices dead.

14.2 EtherNet/IP and CIP

The problem. Rockwell equipment needs cyclic I/O exchange with guaranteed update rates, not just polling.

How it works. Explicit messaging (configuration, non-time-critical) uses TCP port 44818. Implicit messaging — the cyclic I/O data — uses UDP port 2222 and is frequently **multicast**.

Network implication. That multicast is the important part. Without IGMP snooping and a querier the I/O traffic floods every port on the VLAN. This is the most common EtherNet/IP commissioning problem and it presents as unrelated devices becoming slow.

14.3 PROFINET

The problem. Siemens equipment needs deterministic cyclic exchange.

How it works. Three classes. **TCP/IP** for configuration. **RT** (real-time) writes frames directly into Ethernet with EtherType `0x8892`, skipping IP entirely for lower latency. **IRT** (isochronous real-time) adds hardware scheduling for motion control.

Network implication. RT frames have no IP header and therefore **cannot be routed** — controller and devices must share a broadcast domain. IRT requires switches with specific hardware support; ordinary managed switches will not do. PROFINET commonly uses MRP for ring redundancy.

14.4 EtherCAT

The problem. Motion control needs update cycles measured in tens of microseconds.

How it works. A single frame passes through every device in a chain. Each device reads its own data and writes its reply *as the frame passes through*, without storing and forwarding it. One frame serves the whole network per cycle.

Network implication. EtherCAT is not switched Ethernet despite using the same cables. It needs its own dedicated segment and does not mix with ordinary traffic. Treat it as a separate network that happens to use familiar hardware.

14.5 DNP3 and IEC 60870-5-104

The problem. Send plant data to a control centre over a link that may be slow or unreliable.

	DNP3	IEC 60870-5-104
Port	TCP 20000	TCP 2404
Common in	North America	Europe, Asia, India
Reporting	Unsolicited responses	Spontaneous transmission
Security	DNP3-SA adds authentication	IEC 62351 adds security

Both carry time-stamped events, so the control centre's view of event order depends entirely on your plant clocks being correct. Both should terminate on a gateway rather than reaching the plant network directly.

THE POINT LIST IS THE DELIVERABLE

Address numbering, scaling and data types must match what the control centre expects exactly. Mismatches are invisible from the plant side — everything looks healthy locally — and are the most common cause of delay on a control centre interface. Treat the point list as a controlled document.

14.6 IEC 61850

The problem. Electrical substations need protection signals delivered in milliseconds and a common data model across vendors.

Service	Transport	Purpose
MMS	TCP 102	SCADA client/server — reading data, issuing commands

Service	Transport	Purpose
GOOSE	EtherType 0x88B8	Peer-to-peer trip and interlock signals, 4 ms budget
Sampled Values	EtherType 0x88BA	Digitised current and voltage waveforms, continuous

Network implication. GOOSE and Sampled Values are raw Layer 2 multicast with no IP — they cannot be routed, and publishers and subscribers must share a VLAN. Sampled Values produce several megabits per second continuously per stream, so filtering is mandatory. The 4 ms GOOSE budget is why RSTP is unsuitable for a protection network and PRP or HSR is used instead.

14.7 OPC UA and MQTT

	OPC UA	MQTT
Port	TCP 4840	TCP 1883, or 8883 with TLS
Pattern	Client/server, or publish/subscribe	Publish/subscribe through a broker
Carries	Data plus a full information model	Small messages, structure is up to you
Security	Built in — certificates, encryption	Relies on TLS and broker configuration
Use for	Plant to enterprise integration	Telemetry, remote sites, cloud

These are the protocols that carry plant data upward rather than sideways. Both belong at the boundary — pushing to a DMZ — rather than reaching directly into the control network.

SECTION 15

The Physical Layer

15.1 Copper, fibre and optics

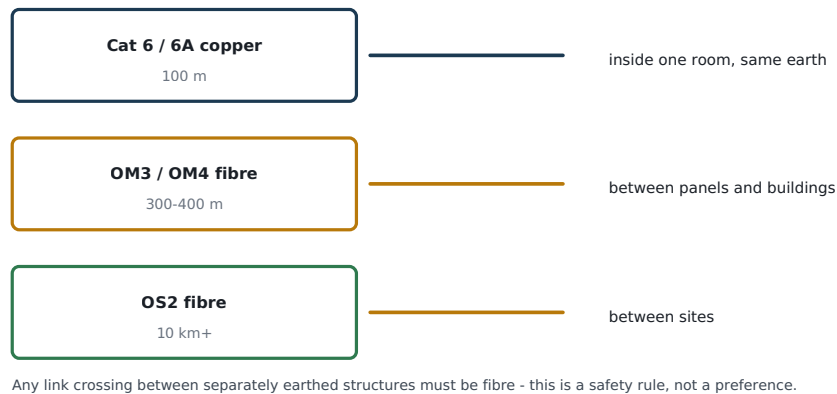


Figure 11. Media selection by distance and environment.

Type	Reach	Notes
Cat 6 / 6A	100 m	Includes patch leads. Shielded where noise is a concern, and the shield must be bonded.
OM3 multimode	300 m at 1 G	Within a building
OM4 multimode	400 m at 1 G	Larger sites
OS2 singlemode	10 km and beyond	Between sites. Specify this if the run might ever grow.

SFP modules let one switch port take copper or any fibre type. Match the module to the fibre and to the far end, and keep spares — a failed optic is faster to swap than to diagnose.

EARTH POTENTIAL RISE

During a fault, the potential difference between two separately earthed structures can reach kilovolts. A copper Ethernet cable between them becomes a conductor carrying that difference, destroying equipment at both ends and creating a personnel hazard.

Any link crossing between separately earthed structures must be fibre. This is a safety requirement, not a performance preference.

15.2 PoE, SPE and APL

Technology	Provides	Use
PoE (802.3af)	15 W	Small cameras, access points
PoE+ (802.3at)	30 W	Pan-tilt-zoom cameras, panel HMIs

Technology	Provides	Use
PoE++ (802.3bt)	60-90 W	Larger devices
Single Pair Ethernet	10 Mbps to 1 km	Replacing 4-20 mA field wiring
Ethernet-APL	SPE plus intrinsic safety	Hazardous areas — process instruments

SPE and APL matter because they extend Ethernet to instruments that previously could not have it — long runs on two wires, and in APL's case into explosive atmospheres. Worth knowing about for new process plant design; not something to retrofit.

APPENDIX

Quick Reference

Ports and identifiers

Protocol	Port or EtherType	Transport
SSH	22	TCP
DNS	53	UDP
DHCP	67, 68	UDP
IEC 61850 MMS	102	TCP
NTP	123	UDP
SNMP / traps	161 / 162	UDP
PTP event / general	319 / 320	UDP
Modbus TCP	502	TCP
Syslog	514	UDP
MQTT / MQTT over TLS	1883 / 8883	TCP
EtherNet/IP implicit I/O	2222	UDP
IEC 60870-5-104	2404	TCP
OPC UA	4840	TCP
DNP3	20000	TCP
EtherNet/IP explicit	44818	TCP
PROFINET RT	0x8892	Ethernet
EtherCAT	0x88A4	Ethernet
GOOSE	0x88B8	Ethernet
Sampled Values	0x88BA	Ethernet
PTP over Layer 2	0x88F7	Ethernet
LLDP	0x88CC	Ethernet

Standards

Standard	Covers
IEEE 802.1Q	VLANs and tagging
IEEE 802.1w / 802.1s	RSTP / MSTP
IEEE 802.1AX (802.3ad)	Link aggregation, LACP
IEEE 802.1AB	LLDP
IEEE 802.1X	Port-based authentication

Standard	Covers
IEEE 802.1AS / Qbv / Qbu / CB	TSN
IEEE 1588-2019	PTP
IEEE C37.238	PTP Power Profile
IEEE 1613	Networking equipment in substations
IEC 61850	Substation communication
IEC 61850-3	Environmental and EMC requirements
IEC 62439-2	MRP
IEC 62439-3	PRP and HSR
IEC 62443	Industrial control system security
RFC 5798	VRRP

Protocols by recovery time

Protocol	Recovery	Topology
PRP / HSR	0 ms	Two LANs / ring
DLR	~3 ms	Ring
MRP fast	10–30 ms	Ring
REP	50–100 ms	Ring or segment
MRP standard	200 ms	Ring
LACP member loss	sub-second	Bundled links
VRRP / HSRP	sub-second to 3 s	Gateway pair
RSTP / MSTP	1–3 s	Any
OSPF	seconds	Routed
STP (legacy)	30–50 s	Any